

The empirical recurrence for OEIS A233660: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A233660 records “Empirical recurrence of order 74”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 343$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 74, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 74 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233660 is “Number of $(n+1) \times (2+1)$ 0..6 arrays with every 2×2 subblock having the sum of the squares of the edge differences equal to 26.”. It has offset 1 and begins

1220, 8324, 56212, 411848, 3007000, 22597004, 168812492, 1277805668, ...

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:56:04, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 343$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 343$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 686$ exact terms of the model returns order 74 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	14	c_{20}	1852086216432408	c_{39}	339765467805252679112	c_{58}	−28712225036184
c_2	136	c_{21}	−2351899978249415	c_{40}	−531167687759830744768	c_{59}	−20101611930472
c_3	−2727	c_{22}	−13453548304568276	c_{41}	−347023538956882436378	c_{60}	82363537978885
c_4	−4717	c_{23}	24950179779783010	c_{42}	807974368494383170446	c_{61}	17912079908053
c_5	234462	c_{24}	75806051779856657	c_{43}	206925514093988691701	c_{62}	−1304801007292
c_6	−261589	c_{25}	−190856570288378500	c_{44}	−954484091093721010768	c_{63}	−138371062385
c_7	−11771403	c_{26}	−324458672689182870	c_{45}	32762246927831673476	c_{64}	1161752706910
c_8	32847963	c_{27}	1118877282432811286	c_{46}	871719919903662735344	c_{65}	−128163621809
c_9	383585273	c_{28}	990331031301479348	c_{47}	−236975982031582331485	c_{66}	−519861005528
c_{10}	−1602467647	c_{29}	−5160543931498593783	c_{48}	−604786979243083565109	c_{67}	108971463958
c_{11}	−8463487806	c_{30}	−1674396588723546875	c_{49}	299706654905918789663	c_{68}	795583761907
c_{12}	48775063190	c_{31}	18953052087982328113	c_{50}	306878475973745252197	c_{69}	−34387318663
c_{13}	125389845250	c_{32}	−1886416235479145322	c_{51}	−232186309000206882899	c_{70}	10688527232
c_{14}	−1042146512052	c_{33}	−55641434075762975085	c_{52}	−104148533262552395666	c_{71}	3606546022
c_{15}	−1103910217676	c_{34}	25090355630858008662	c_{53}	124246235483306346712	c_{72}	−309122457
c_{16}	16481866449739	c_{35}	130183180246855610069	c_{54}	16791624448672887033	c_{73}	41107456
c_{17}	1172660306786	c_{36}	−100902434493108042036	c_{55}	−46746108002822185445	c_{74}	688128
c_{18}	−198588905869757	c_{37}	−240012139720332542737	c_{56}	3400073316860065911		
c_{19}	131017445695181	c_{38}	268632788511530741708	c_{57}	12086554193806852085		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 74, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $74 + 4$ terms removed returns order 74 as well, so $\deg q_{\min} = 74 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $74 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 74 that this sequence satisfies — and that family turns out to have exactly one member.

References

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